

Ankle pain

with KAREN HOLZER and JON KARLSSON

Dance is the hidden language of the soul.

Martha Graham, dancer and choreographer, 1894–1991

Sportspeople, particularly ballet dancers, footballers, basketballers and high jumpers, may complain of ankle pain that is not related to an acute ankle injury (Chapter 41). Clinical management of such 'overuse' injuries is simplified if the presentations are further divided into:

- medial ankle pain
- lateral ankle pain
- anterior ankle pain
- posterior ankle pain: defined in this book as the 'Achilles region' (Chapter 40).

It is important to note that clinically, patients often present with combinations of pain, such as anterolateral ankle pain in soccer players. In those circumstances, the systematic clinical approach outlined in this chapter still aids in diagnosis and management.

MEDIAL ANKLE PAIN

Although there are numerous causes of medial ankle pain, the most common is tendinopathy, in particular of the tibialis posterior tendon, and to a lesser degree the flexor

hallucis longus tendon. Another important cause of medial ankle pain is tarsal tunnel syndrome, where the posterior tibial nerve is compressed behind the medial malleolus. This may present as medial ankle pain with sensory symptoms distally. Posterior impingement syndrome of the ankle can also present as medial ankle pain (Chapter 40).

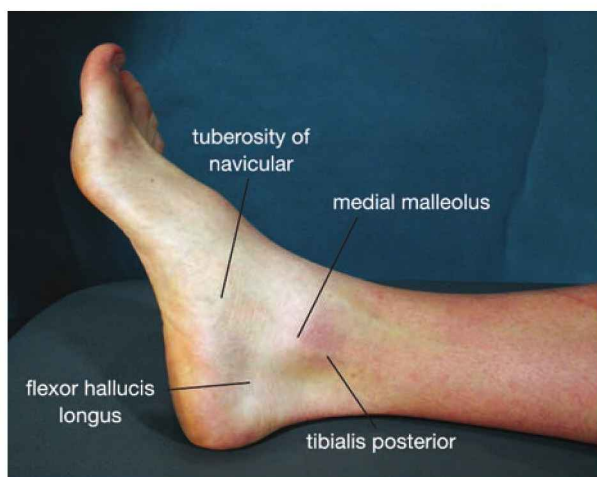
The causes of medial ankle pain are listed in Table 42.1. The anatomy of the region is illustrated in Figure 42.1.

History

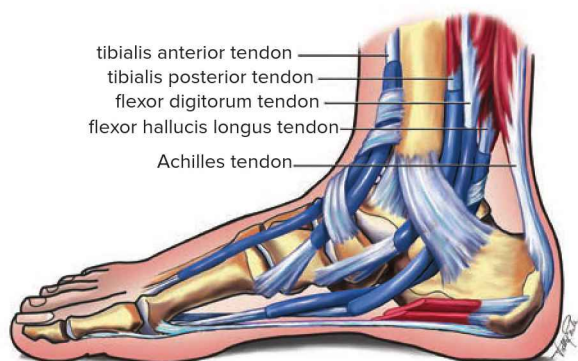
In patients with medial ankle pain there is usually a history of overuse, especially running or excessive walking (tibialis posterior tendinopathy), toe flexion in ballet dancers and high jumpers (flexor hallucis longus tendinopathy) or plantarflexion in dancers and footballers (posterior impingement syndrome). In the case of tibialis posterior tendinopathy, pain may radiate along the line of the tendon to its insertion on the navicular tubercle, or in tarsal tunnel syndrome the arch of the foot. Associated sensory symptoms such as pins and needles or numbness

Table 42.1 Causes of medial ankle pain

Common	Less common	Not to be missed
Tibialis posterior tendinopathy Flexor hallucis longus tendinopathy	Medial calcaneal nerve entrapment Calcaneal stress fracture Tarsal tunnel syndrome Talar stress fracture Medial malleolar stress fracture Posterior impingement syndrome (Chapter 40) Referred pain from lumbar spine	Navicular stress fracture (Chapter 43) Complications of acute ankle injuries (Chapter 41) Complex regional pain syndrome type 1 (following knee or ankle injury) (Chapter 5)



(a)



(b)

Figure 42.1 Medial aspect of the ankle (a) Surface anatomy (b) Anatomy of the medial ankle

may suggest tarsal tunnel syndrome. Crepitus is commonly associated with flexor hallucis longus and occasionally tibialis posterior tendinopathy.

Examination

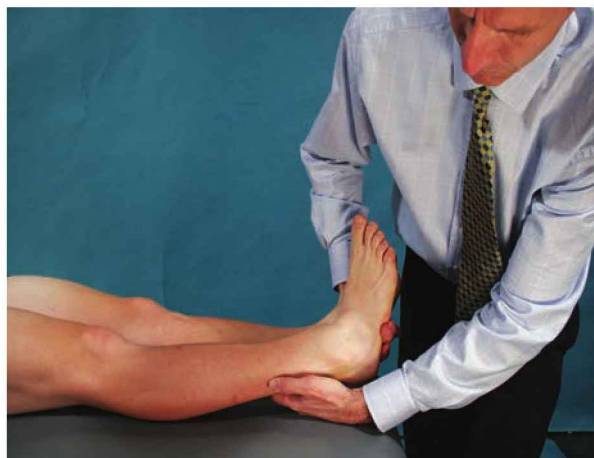
Careful palpation and testing of resisted movements is the key to examination of this region.

1. Observation
 - a. standing
 - b. walking
 - c. supine
2. Active movements
 - a. ankle plantarflexion/dorsiflexion
 - b. ankle inversion/eversion
 - c. flexion of the first metatarsophalangeal joint

3. Passive movements
 - a. as for active
 - b. subtalar joint
 - c. midtarsal joint
 - d. muscle stretches
 - i. gastrocnemius
 - ii. soleus
4. Resisted movement
 - a. inversion (Fig. 42.2a)
 - b. first toe flexion; check for crepitation (Fig. 42.2b)
5. Functional tests
 - a. hop
 - b. jump
6. Palpation
 - a. tibialis posterior tendon (Fig. 42.2c)
 - b. flexor hallucis longus; behind the medial malleolus
 - c. navicular tubercle
 - d. ankle joint
 - e. midtarsal joint
 - f. Achilles tendon
7. Special tests
 - a. Tinel's test (Fig. 42.2d)
 - b. sensory examination; especially for nerve entrapment of the calcaneal branches (Baxter's nerve) (Fig. 42.2e)
8. biomechanical examination (Chapter 8)
9. lumbar spine examination (Chapter 29).

Key outcome measures

The key patient-reported outcome measures (PROMs) used in ankle pain are shown in Table 42.2.



(a)

Figure 42.2 Examination of the patient with medial ankle pain (a) Resisted movement—inversion (tibialis posterior)



(b)



(d)



(c)



(e)

Figure 42.2 (cont.) (b) Resisted movement—Toe flexion (flexor hallucis longus) (c) Palpation—tibialis posterior tendon. The tibialis posterior tendon is palpated from posteromedial to the medial malleolus to its insertion at the navicular tubercle (d) Special tests—Tinel's test. Tapping over the posterior tibial nerve in the tarsal tunnel may reproduce symptoms (e) Special tests—sensory examination especially for nerve entrapment of the calcaneal branches (Baxter's nerve). (See also Chapter 43, and refer to Figure 43.7.)

Investigations

Plain radiograph is indicated when the following conditions are suspected:

- Posterior impingement: to confirm the presence of either a large posterior process of the talus/fracture of the posterior process of the talus or an os trigonum. A lateral view with the foot in a maximally plantarflexed position (posterior impingement view) can be useful to determine if bony impingement is occurring.
- Anteromedial impingement: anteroposterior views may reveal osteophytes with joint narrowing between the

medial talus and medial malleolus, or the anteromedial aspect of the medial tibial plafond.

- Stress fracture of the medial malleolus: often a delayed diagnosis, and as such sclerosis and/or a fracture line may be seen.

Ultrasound can assist in the diagnosis of tibialis posterior or flexor hallucis longus tenosynovitis or tendinopathy. MRI assists in the diagnosis of most medial ankle conditions including impingements, tenosynovitis and tendinopathies, and bony injuries.

Nerve conduction studies may be performed if tarsal tunnel syndrome is suspected. However nerve conduction studies are seldom diagnostic.

Table 42.2 Patient-reported outcome measures for the ankle**Generic foot and ankle pain measures*****Foot and Ankle Ability Measure (FAAM)¹***

- Self-administered questionnaire evaluating difficulty with activities of daily living (ADL) and sporting activities for patient with foot and ankle pain
- Two subscales: ADL and Sports subscales
- 29 items (21 on difficulty with ADL, 8 on difficulty with sporting activities)
- Score range: 0–100 for each subscale
- Test-retest reliability:
 - ADL subscale: 0.89
 - Sports subscale: 0.87
- Construct validity:
 - ADL subscale: SF-36¹ PF subscale and PCS subscale: both 0.84¹
 - Higher ADL subscale scores for healthy athletes (M=100, SD=0) compared to patients with chronic ankle instability (M=88, SD=7.7)
 - Sports subscale: SF-36¹ PF subscale and PCS subscale: 0.78 and 0.80 respectively¹
 - Higher Sports subscale scores for healthy athletes (M=99, SD=3.5) compared to patient with chronic ankle instability (M=76, SD=12.6)
- Minimal detectable change (MCD95%):
 - ADL subscale: 5.7
 - Sports subscale: 12.3
- Minimal clinically important difference (MCID):
 - ADL subscale: 8
 - Sports subscale: 9* (* as determined with optimal sensitivity/specificity in ROC curve)
- Developed and validated in American English, cross-culturally validated to Brazilian Portuguese,³ Dutch,⁴ German,⁵ Italian,⁶ Japanese,⁷ Korean,⁸ Persian,⁹ Thai¹⁰ and Turkish¹¹

Oxford Ankle Foot Questionnaire^{12, 13}

- Self-administered questionnaire evaluating disability in children with foot and ankle problems (5–16 years of age)
- 15 items (6 on physical domain, 4 on school and play domain, 4 on emotional domain and 1 on footwear)
- Score range:
 - Physical domain: 0–24 (0–100 if transformed to)
 - School and Play domain: 0–16 (0–100 if transformed to %)
 - Emotional domain: 0–16 (0–100 if transformed to %)
- Test-retest reliability:
 - Physical domain: 0.62¹² and 0.85¹³
 - School and Play domain: 0.87 and 0.95
 - Emotional domain: 0.83 and 0.78
 - Footwear item: 0.76¹³
- Construct validity¹²
 - Physical domain: Kidscreen physical wellbeing: 0.60
 - School and Play domain: Kidscreen physical wellbeing and school: 0.54 and 0.31 respectively
 - Emotional domain: Kidscreen psychological wellbeing: 0.46
 - Footwear item: N/A
- MDC90%:
 - Physical domain: 7.1¹³
 - School and Play domain: 5.4¹³
 - Emotional domain: 8.0¹³

Oxford Ankle Foot Questionnaire (cont)

- MCID:
 - Physical domain: 10.0¹³
 - School and Play domain: 8.7¹³
 - Emotional domain: 8.3¹³
- Developed and validated in British English,^{12–14} cross-culturally validated to Danish¹⁵ and Italian¹⁶
- Available online at www.orthopaedicscore.com/scorepages/foot_and_ankle_disability_index_fadi.html

Foot and Ankle Disability Index (FADI) Score

- Self-administered questionnaire that evaluated ADL and sporting activities
- 34 items (26 on ADL and 8 on sports)
- Score range:
 - FADI ADL 0–104¹⁷
 - FADI Sport 0–32¹⁷
 - Score transformation to 0–100%; 100% means no dysfunction
- Test-retest reliability:
 - FADI ADL: 0.89 and 0.93¹⁷
 - FADI Sport: 0.84 and 0.92¹⁷
- Developed¹⁸ and validated¹⁷ in American English
- Available online at www.orthopaedicscore.com/scorepages/foot_and_ankle_disability_index_fadi.html

American Academy of Orthopaedic Surgeons (AAOS) Foot and Ankle Module

- Self-administered questionnaire evaluating symptoms and function of the foot and ankle
- 25 items (20 global foot ankle scale; 5 items shoe comfort scale)
- Scores are transformed to 0–100% for each scale; 100% means no dysfunction
- Test-retest reliability; global rating scale, $r = 0.79$; shoe comfort scale, $r = 0.87$ ¹⁹
- Construct validity:
 - Foot and Ankle Module (both subscales): FAAM 0.92,²⁰ AOFAS 0.81,²⁰ SF-12 Physical 0.69,²⁰ Foot function index -0.85²⁰
 - Global Foot and Ankle Scale: physician-assessed function: 0.43;¹⁹ physician-assessed pain: 0.49;¹⁹ SF-36 Physical Health: 0.65¹⁹
- Developed and validated in American English,¹⁹ cross-culturally validated to Spanish²¹ and Korean²²

American Orthopaedic Foot & Ankle Society (AOFAS) Clinical Rating System

- Clinician-administered, region-specific questionnaires evaluating pain, function and alignment in patients with foot and ankle pain
- Four separate scales:²³
 - Ankle-Hindfoot Scale (9 items)
 - Midfoot Scale (7 items)
 - Hallux Metatarsophalangeal-Interphalangeal Scale (8 items)
 - Lesser Metatarsophalangeal-Interphalangeal Scale (8 items)
- 0–100 points; 100 = no dysfunction, for all scales
- Test-retest reliability:
 - Ankle-Hindfoot Scale: 0.71²⁴
 - Midfoot Scale: N/A
 - Hallux Metatarsophalangeal-Interphalangeal Scale: 0.95²⁵ and 0.82²⁴
 - Lesser Metatarsophalangeal-Interphalangeal Scale: 0.80²⁵
- Construct validity:
 - Ankle-Hindfoot Scale:
 - SF-36: Physical component scale: 0.27;²⁶ Bodily pain: 0.53;²⁶ Physical functioning: 0.34;²⁶ Vitality 0.41²⁶

continued

Table 42.2 *Cont.***Generic foot and ankle pain measures (cont)*****American Orthopaedic Foot & Ankle Society (AOFAS) Clinical Rating System***

- Manchester-Oxford Foot Questionnaire: Walking/standing: -0.56 ;²⁷ Pain: -0.60 ²⁷
- Generic Health quality-adjusted life-year (QUALY) Score: $0.24-0.47$ ²⁸
- Self-reported Foot and Ankle Score: 0.49 ²⁴
- Midfoot Scale:
 - Generic Health QUALY Score: $0.23-0.37$ ²⁸
- Hallux Metatarsophalangeal-Interphalangeal Scale:
 - Foot Function Index: -0.81 ²⁵
 - Generic Health QUALY Score: $0.35-0.58$ ²⁸
 - Self-reported Foot and Ankle Score: 0.67 ²⁴
- Lesser Metatarsophalangeal-Interphalangeal Scale
 - Generic Health QUALY Score: $0.22-0.67$ ²⁸
- MDC 90%:
 - Ankle-Hindfoot Scale: 6.30 ²⁹
 - Midfoot Scale: 6.74 ²⁹
 - Hallux Metatarsophalangeal-Interphalangeal Scale: 7.10 ²⁹
 - Lesser Metatarsophalangeal-Interphalangeal Scale: 7.12 ²⁹
- MCID:
 - Ankle-Hindfoot Scale: 8.90 ²⁹
 - Midfoot Scale: 12.40 ²⁹
 - Hallux Metatarsophalangeal-Interphalangeal Scale: 24.75 ²⁹
 - Lesser Metatarsophalangeal-Interphalangeal Scale: 11.33 ²⁹
- Developed and validated in American English,^{23-26, 28, 29} cross-culturally validated to Brazilian Portuguese,^{30, 31} German³² and Italian³³

Foot and Ankle Outcome Score

- Self-administered questionnaire evaluating pain, symptoms, ADL, sport and recreation function, and foot and ankle-related quality of life (QOL)
- 42 items (9 on pain, 7 on symptoms, 17 on ADL, 5 on Sport and Recreation and 4 on foot and ankle QoL)
- Score range 0–100 (100 = no problem)
- Test-retest reliability:
 - Pain: 0.78 ³⁴
 - Symptoms: 0.86 ³⁴
 - ADL: 0.70 ³⁴
 - Sport and recreation: 0.85 ³⁴
 - Foot and Ankle QoL: 0.92 ³⁴
- Construct validity: Karlsson Score $0.58-0.67$ ³⁴
- Developed and validated in Swedish, cross-culturally validated to Dutch,³⁵⁻³⁷ German,³⁸ Korean,³⁹ Persian⁴⁰ and Turkish⁴¹
- Validated in American English, without cross-cultural translation process^{21, 42-45}
- Non-validated versions available in Arabic, Brazilian Portuguese, Chinese, Danish, Norwegian, Polish and Portuguese
- Available online in multiple languages: www.koos.nu

Lower limb pain measures***Lower Extremity Functional Score (LEFS)***

- Self-administered questionnaire: difficulty with various activities
- 20 items
- Score ranges from 0–80 (80 = no difficulty)⁴⁶
- Test-retest reliability: intraclass correlation coefficient (ICC) = $0.87-0.96$ ^{46, 47}
- Construct validity: SF-36 physical function: $r = 0.80$; SF-36 physical component: $r = 0.80$;⁴⁶ Olerud Molander Ankle Score: $r = 0.80-0.87$,⁴⁸ LLFI: 0.88 ⁴⁹

Lower limb pain measures (cont)

- MDC 90%: 9,⁴⁶ 9.4,⁴⁷ 13;⁵⁰ MDC 95%: 15⁵⁰
- MCID: 9⁴⁶
- Developed and validated in American English,⁴⁶ cross-culturally translated and validated in Arabic,⁵¹ Brazilian Portuguese,⁵² Dutch,⁵³ Finnish,⁵⁴ French,⁵⁵ Italian,⁵⁶ Persian,⁵⁷ Spanish,⁵⁸ Taiwanese⁵⁹ and Turkish⁶⁰

Lower Limb Functional Index (LLFI)

- Self-administered questionnaire evaluating lower limb function
- 25 items
- Score ranges from 0–100 (100 = no difficulty)
- Test-retest reliability: ICC = 0.97⁴⁹
- Construct validity: LEFS: 0.88⁴⁹
- MDC 90%: 6.63⁴⁹
- Developed and validated in a mixed Australian/American population,⁴⁹ cross-culturally translated and validated in Spanish⁶¹ and Turkish⁶²

Lower Limb Task Questionnaire

- Self-administered questionnaire evaluating ADL and recreational activities
- Two subscales, both scores range from 0–40 (40 = no difficulty)
- Test-retest reliability:
 - ADL, ICC = 0.96⁶³
 - Recreational activities, ICC = 0.98⁶³
- Construct validity:
 - ADL: Motor activity scale, $r = 0.62^{63}$
 - Recreational activities: Motor activity scale, $r = 0.72^{63}$
- MDC95%: N/A
- MCID:
 - ADL: 10.1⁶³
 - Recreational activities: 7.7⁶³
- Developed and validated in New Zealand⁶³

Ankle Joint Functional Assessment Tool

- Self-administered questionnaire evaluating ankle function
- Score ranges from 0–48 (48 = full function)
- Developed in American English; no validation studies were found

Tibialis posterior tendinopathy

The tibialis posterior tendon is the largest and most anterior structure that passes behind the medial malleolus (Fig. 42.1b). It then divides and sends attachments to the navicular tuberosity, the cuboid, cuneiforms, bases of the second to fourth metatarsals and the spring ligament.

The functions of the tibialis posterior include:

- inversion of the subtalar joint
- dynamic stabilisation of the hindfoot against valgus (eversion) forces
- support of the medial longitudinal arch
- locking the transverse tarsal joints, thus allowing the gastrocnemius to support heel raise, and assisting the Achilles tendon in plantarflexion of the ankle.

Causes

Acute injuries of the tibialis posterior tendon are rare. Thus, tibialis posterior tendinopathy is usually related to overuse and is most often seen in middle-aged women, and in ballet dancers. It comes on slowly, and is not self-limiting.

1. Overuse, often related to:
 - a. excessive walking, running or jumping
 - b. excessive subtalar pronation. This increases eccentric tendon loading during supination for toe-off
2. Acute:
 - a. direct trauma–laceration
 - b. indirect trauma–eversion ankle sprain, ankle fracture
 - c. acute avulsion fracture

3. Inflammatory conditions:

- a. tenosynovitis secondary to rheumatoid arthritis and seronegative arthropathies.

Chronic tendinopathies are characterised by collagen disarray and interstitial tears, and may eventually lead to total tendon rupture.

Clinical features

The clinical features of tibialis posterior tendinopathy are as follows.

- Medial ankle pain often arising from behind the medial malleolus and extending towards the medial midfoot to the insertion of the tendon at the navicular bone.
- Swelling is very unusual in the early stages and if present may suggest a substantial tendon injury or an underlying seronegative arthropathy. In the later stages, however, swelling is more pronounced and is often diffuse around and below the medial malleolus.
- Tenderness along the tendon is usual, most prominent posterior and inferior to the medial malleolus or at the insertion onto the navicular bone.
- Crepitus is only occasionally present.
- Resisted inversion (Fig. 42.2a) will elicit pain and relative weakness compared with the contralateral side.
- A single heel raise test viewed from behind will reveal lack of inversion of the hindfoot, and if severe the patient may have difficulty performing a heel raise. In the normal ankle, the calcaneus moves into varus position during a single heel raise; in patients with tibialis posterior tendinopathy, this does not happen and the heel remains in the valgus position.

Investigations

MRI (Fig. 42.3) or ultrasound may confirm the diagnosis and reveal the extent of tendinosis, including the presence of tears. MRI is the imaging modality of choice for the evaluation of tibialis posterior tendinopathy dysfunction. MRI has a sensitivity of 95%, a specificity of 100% and an accuracy of 96% in identifying disorders of the tendon.⁶⁴ It also has the ability to recognise injuries associated with longer-term tibialis posterior dysfunction, such as spring ligament failure, sinus tarsi inflammation and midfoot degeneration.

Treatment

In milder cases conservative treatment is recommended; however, in more severe cases surgery may be required. Frequently the patient presents with an acute painful tendon, termed a 'reactive tendinopathy' which should be treated with rest, ice and non-steroidal anti-inflammatory drugs (NSAIDs) in the initial phase. Once the pain

and inflammation has settled, treatment includes the following.

- Deep soft tissue therapy to the tibialis posterior muscle.
- Footwear: a stability shoe is recommended and often the wearing of an orthoses to control excessive pronation.
- Progressive concentric and eccentric tendon loading exercise program once the inflammation has settled (Fig. 42.4).

PRACTICE PEARL

Corticosteroid injections should never be used in the management of tibialis posterior tendinopathy due to poor long-term outcomes and an increased risk of total tendon rupture.

The use of platelet rich plasma (PRP) injections has no biological rationale and no trial evidence (see also Chapter 17).

Surgery is recommended if there is tendon rupture (Chapter 41) or if conservative management fails to settle the condition. In the case of tenosynovitis, a synovectomy may be performed, while in cases of severe tendinopathy or tendon rupture a reconstruction may be required. There is often need for large corrective surgical treatment. This treatment is directed towards correct alignment of the heel, and may include tendon transfer, calcaneus osteotomy, and/or midfoot corrective osteotomy/arthrodesis.

One major problem with the management of tibialis posterior tendinopathy is the long time from injury to final treatment which can result from delayed presentation or delay in diagnosis.

Flexor hallucis longus tendinopathy

The flexor hallucis longus (FHL) tendon flexes the big toe and assists in plantarflexion of the ankle. It passes posterior to the medial malleolus, and runs between the two sesamoid bones to insert into the base of the distal phalanx of the big toe.

Causes

FHL tendinopathy may occur secondary to overuse, a stenosing tenosynovitis, a pseudocyst or tendon tear. The tendon tear may be partial or total. A common cause is overuse in a ballet dancer, as dancers repetitively go from flat-foot stance to the *en pointe* position when extreme plantarflexion is required. Wearing shoes that are too big and require the athlete to 'toe-grip' may also result in flexor hallucis longus tendinopathy.

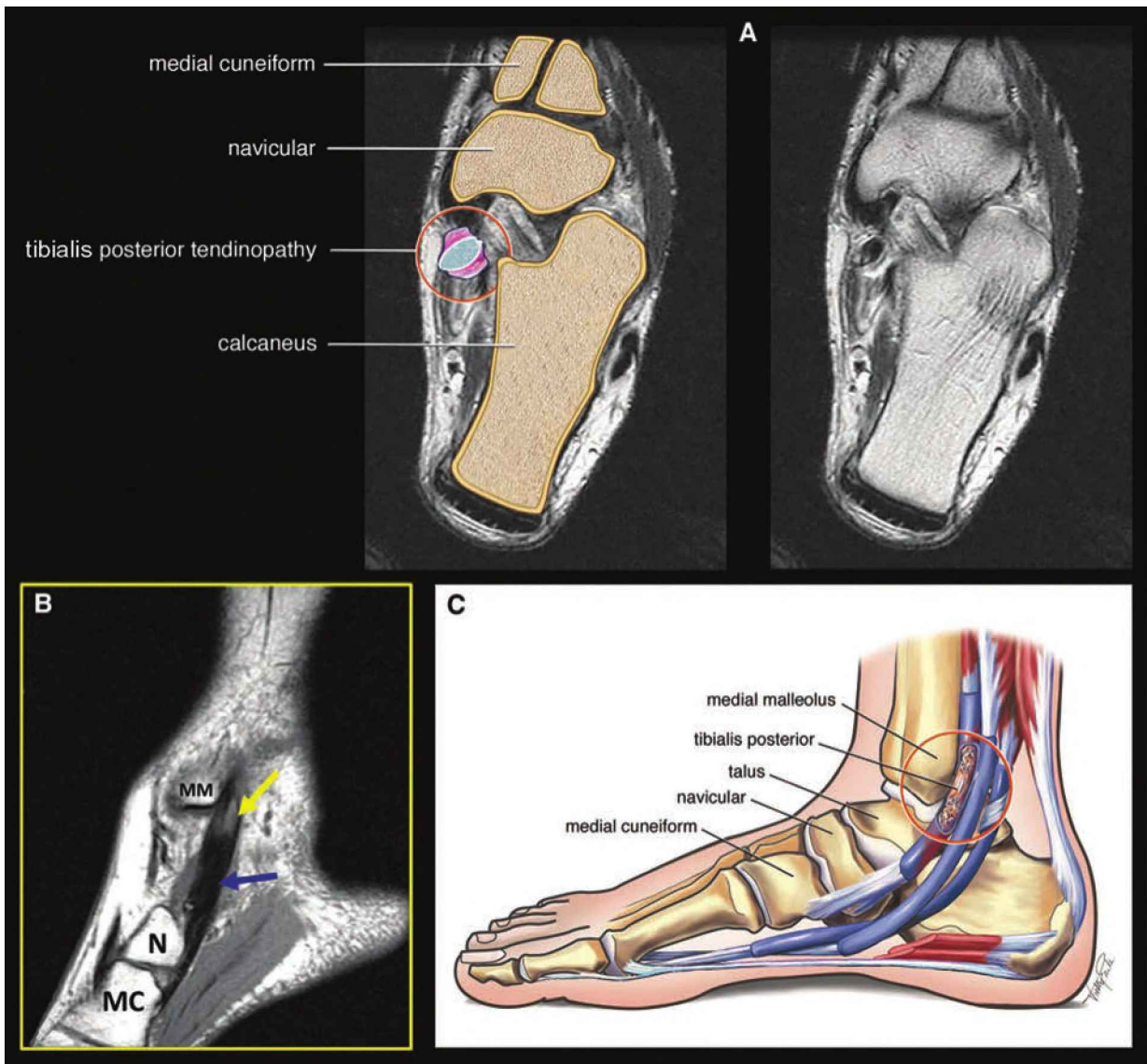


Figure 42.3 Tibialis posterior tendinopathy (a) Axial view showing tendon surrounded by area of oedema (b) Sagittal view showing abnormal (yellow arrow) and normal (blue arrow) tendon appearance (c) Anatomy

This condition is often associated with posterior impingement syndrome (Chapter 40) as the flexor hallucis tendon lies in a fibro-osseous tunnel between the lateral and medial tubercles of the posterior process of the talus. Enlargement or medial displacement of the posterior process of the talus or os trigonum compresses the flexor hallucis longus at the point where the tendon changes direction from a vertical course dorsal to the talus to a horizontal course beneath the talus (Fig. 42.5). This can cause tendon thickening and may result in 'triggering' of the tendon with loud and painful crepitation, as a consequence of partial tearing and

subsequent healing of the tendon with excessive scar tissue production.

Clinical features

The clinical features of FHL tendinopathy are as follows.

- Pain is on toe-off or forefoot weight-bearing (e.g. rising in ballet), maximal over the posteromedial aspect of the medial malleolus and the calcaneus around the sustentaculum tali.
- Pain landing from jumps (key differentiator from posterior impingement syndrome)



Figure 42.4 Eccentric exercises—tibialis posterior. Patient stands on the edge of a step and drops down into eversion, eccentrically contracting the tibialis posterior muscle

- Pain may be aggravated by resisted flexion (Fig. 42.2b) or passive stretch into full dorsiflexion of the hallux.
- Prominent crepitus in the sheath of flexor hallucis longus
- In more severe cases, there may be 'triggering' of the first toe, both with rising onto the ball of the foot (e.g. in ballet) and in lowering from this position. Triggering occurs when the foot is placed in plantarflexion and the athlete, unable to flex the hallux, is able to extend the interphalangeal or metatarsophalangeal joints of the toe with forcible active contraction of the FHL. A snap or pop (crepitus) occurs in the posteromedial aspect of the ankle when this happens. Subsequent passive flexion or extension of the interphalangeal joint produces a painless snap posterior to the medial malleolus.

Investigations

MRI and ultrasound may both reveal pathology. However, the diagnosis is primarily based on clinical signs as the correlation between imaging findings and symptoms is often limited. The characteristic MRI sign (Fig. 42.6) in the case of tendon pathology, such as partial rupture, is abrupt fluid cut-off in the tendon sheath; excessive fluid is found located around a normal-appearing tendon proximal to the fibro-osseous canal.⁶⁵

Treatment

In the acute phase, treatment may include:⁶⁵

- ice
- avoidance of activities that stress the flexor hallucis longus tendon (e.g. dancer working at the barre but not rising *en pointe*)
- flexor hallucis longus strengthening and stretching exercises

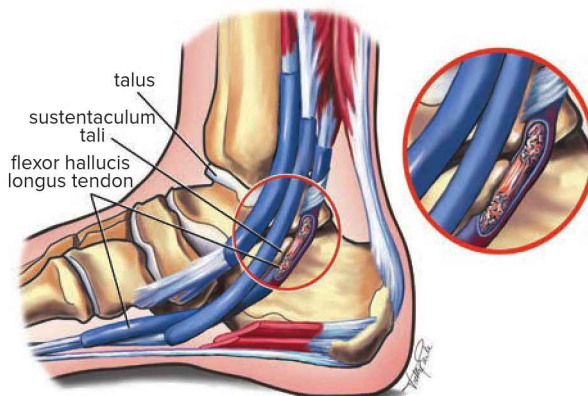


Figure 42.5 Flexor hallucis longus tendinopathy showing the tendon irritated near the medial malleolus

- soft tissue therapy to the proximal muscle belly
- correction of subtalar joint and 1st MTP joint hypomobility with manual mobilisation and orthoses
- control of excessive pronation during toe-off with tape or orthoses—this may be helpful but is difficult to achieve in dancers.

PRACTICE PEARL

To prevent recurrences of FHL tendinopathy, the dancer should focus on slightly reducing the amount of hip turnout thus ensuring that the weight is directly over the hip, avoiding hard floors, and using firm, well-fitting pointe shoes, so that the foot is well supported and no additional strain is placed on the tendon. Technique correction is important in ballet dancers as the tendinopathy arises not only from excessive ankle eversion or inversion with pointe work, but also from proximal weakness, such as poor trunk control.⁶⁶

If symptoms persist after consultation with an expert ballet clinician, surgical treatment should be considered when persistent synovitis or triggering prevents dancing. Arthroscopic surgery involves exploration of the tendon and release of the tendon sheath with a posterior approach. The fibrous band around the tendon is cleaned off and the tendon released. Progressive return to ballet after such surgery takes in the order of 2–3 months.

Tarsal tunnel syndrome

Tarsal tunnel syndrome occurs as a result of entrapment of the posterior tibial nerve in the tarsal tunnel where

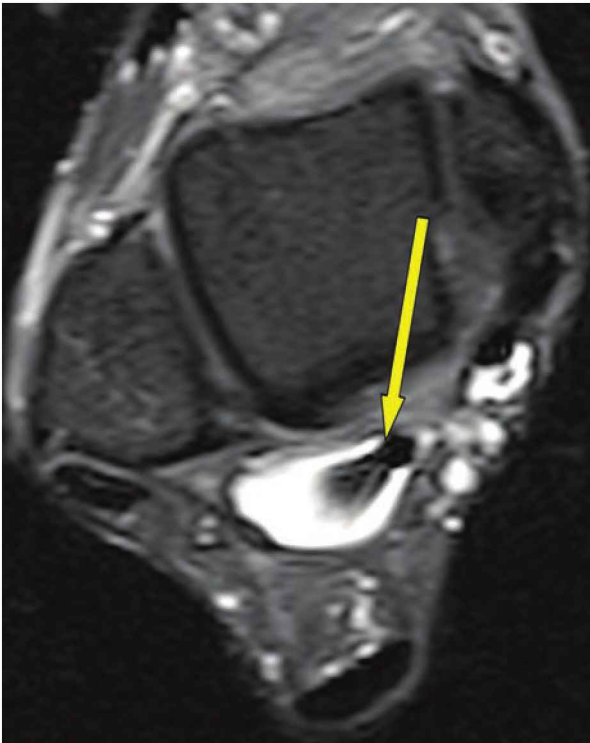


Figure 42.6 Flexor hallucis longus tendinopathy. Axial short tau inversion recovery (STIR) MRI demonstrates a large amount of fluid along the flexor hallucis longus at the level of the myotendinous junction (arrow)

the nerve winds around the medial malleolus.⁶⁷ It may also involve only one of its terminal branches distal to the tarsal tunnel. In such cases, the clinical presentation is pain in the medial part of the heel and sensory disturbance in the same region.

Causes

In approximately 50% of cases, the cause of tarsal tunnel syndrome is idiopathic. It may also occur as a result of trauma (e.g. inversion injury to the ankle, direct blow to the ankle with significant bleeding) or overuse associated with excessive pronation. Other less common causes include:

- ganglion
- talonavicular coalition
- varicose veins
- synovial cyst
- lipoma
- accessory muscle—flexor digitorum accessorius longus
- tenosynovitis; often associated with arthritis (sero-negative rheumatoid arthritis)
- fracture of the distal tibia or calcaneus.

Clinical features

The clinical features of tarsal tunnel syndrome are as follows.

- Poorly defined burning, tingling or numb sensation on the plantar aspect of the foot, often radiating into the toes.
- Pain is usually aggravated by activity and relieved by rest. The location of the pain can be diffuse and poorly defined.
- In some patients, the symptoms are worse in bed at night, and relieved by getting up and moving, or massaging the foot.
- Swellings, varicosities or thickenings may be found on examination around the medial ankle or heel.
- A ganglion or cyst may be palpable in the tendon sheaths around the medial ankle.
- Tenderness in the region of the tarsal tunnel is common.
- Tapping over the posterior tibial nerve (Tinel's sign) may elicit the patient's pain and very occasionally cause fasciculations; however, this 'classic' sign is not commonly seen.
- There may be altered sensation along the arch of the foot.
- The distribution of the sensory changes in the foot needs to be differentiated from the typical dermatomal distribution of S1 nerve root compression.

Investigations

Nerve conduction studies should be performed.^{68, 69} These not only help to confirm the diagnosis but they can also guide the surgeon as to the location of the nerve compression. However, it must be borne in mind that tarsal tunnel syndrome is primarily a clinical diagnosis and nerve conduction studies are negative in approximately 50% of patients. Ultrasound or MRI may be required to assess for a space-occupying lesion as a cause of the syndrome. A plain radiograph and, if required, a computed tomography (CT) scan should be performed in the case of excessive pronation or if a tarsal coalition is suspected.

Differential diagnoses

Differential diagnoses include entrapment of the medial and/or lateral plantar nerves (or both), plantar fasciopathy, intervertebral disc degeneration and other causes of nerve inflammation or degeneration. The diagnosis of tarsal tunnel is often difficult and the symptoms and signs are often vague.⁷⁰ This is important to bear in mind, at least before deciding on surgical intervention.

Treatment

Non-surgical treatment should be attempted in those with either an idiopathic or biomechanical cause. If neurodynamic tests prove positive, then appropriate

gliding treatments are indicated (Chapter 17). If excessive pronation is present, initially taping, and if successful an orthosis, may be required.

Surgical decompression of the posterior tibial nerve and its branches may be indicated in selected cases, but only after both the diagnosis and site of nerve entrapment have been confirmed.

PRACTICE PEARL

Surgery should be reserved as the last resort, if all other treatment fails, symptoms are severe and the clinician is confident of mechanical pressure on the nerve. Results of surgery have not been encouraging,⁷¹ with a high perioperative complication rate.⁷²

Medial malleolar stress fracture

Medial malleolar stress fractures are rare, but should be considered in runners or footballers presenting with persistent medial ankle pain aggravated by activity.⁷³ Although the fracture line is frequently vertical from the junction of the tibial plafond and the medial malleolus, it may arch obliquely from the junction to the distal tibial metaphysis.

Clinical features

Athletes with medial malleolus stress fractures typically present with anterior and medial ankle pain that progressively increases with running and jumping activities. The onset of pain may be acute or gradual. Examination reveals tenderness overlying the medial malleolus, frequently in conjunction with a mild ankle effusion.

Investigations

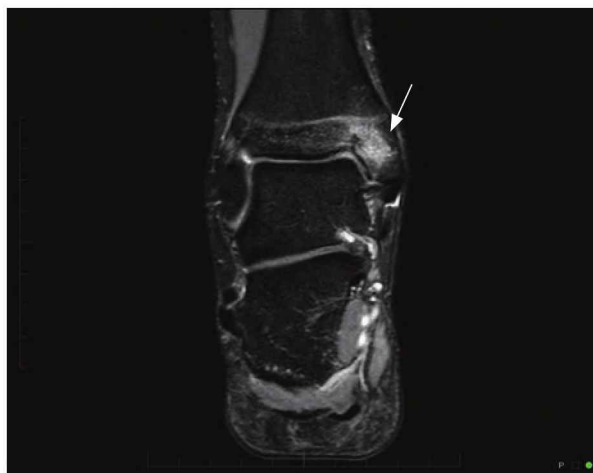
In the early stages, X-rays may be normal, but with time a linear area of hyperlucency may be apparent, progressing to a lytic area and fracture line (Fig. 42.7a). If the X-ray is normal, an MRI is recommended (Fig. 42.7b).

Treatment

For most fractures 4–6 weeks NWB in a CAM boot is required. Following fracture healing, the practitioner should assess biomechanics and footwear. A graduated return to activity follows and usually takes around 12 weeks.⁷⁴ In the case of complete fracture, surgical intervention with internal screw fixation (percutaneous screw fixation) followed by immobilisation in plaster for 3 weeks and air-cast for a further 3 weeks is advisable. A recent systematic review showed that non-operative and



(a)



(b)

Figure 42.7 Medial malleolar stress fracture (a) X-ray showing sclerotic line (arrow) through medial malleolus indicating a stress fracture (b) MRI. Fat suppressed T2-weighted image showing bright T2 signal (arrow) at the site of the stress fracture

operative interventions have proven to be successful with regard to healing and return to play for medial malleolar stress fractures in the recreational and competitive athlete.⁷⁵ However, early operative intervention can possibly create a higher likelihood of early healing, decrease in symptoms, and return to play.

Medial calcaneal nerve entrapment

The medial calcaneal nerve is a branch of the posterior tibial nerve arising at the level of the medial malleolus or below, and passing superficially to innervate the skin of the heel. Occasionally it may arise from the lateral plantar

nerve, a branch of the posterior tibial nerve. It has been theorised that a valgus hindfoot may predispose joggers to compression of this nerve branch. This nerve entrapment is often termed 'Baxter's nerve' (see Chapter 43).

Clinical features

Entrapment or irritation of the nerve causes burning pain over the infero-medial aspect of the calcaneus, which often radiates into the arch of the foot, and is aggravated by running. Examination reveals tenderness over the medial calcaneus and a positive Tinel's sign. There is frequently associated excessive pronation of the hindfoot.

Investigations

The diagnosis may be difficult to establish. Nerve conduction studies can be helpful, and injection of local anaesthetic at the point of maximal tenderness with a resultant disappearance of pain will help confirm the diagnosis.

Treatment

Treatment involves minimising the trauma to the nerve with a change of footwear or the use of a pad over the area to protect the nerve. If this is not successful, injection of corticosteroid and local anaesthetic into the area of point tenderness may be helpful. If this fails to settle, other treatment options include radiofrequency denervation (RFD) of the nerve and surgical decompression.⁷⁶

Other causes of medial ankle pain

Two conditions that generally cause foot pain, but may present as medial ankle pain, are stress fractures of the calcaneus and the navicular bone (Chapter 43). Referred pain from neural structures may occasionally present as medial ankle pain. Entrapment of the medial plantar nerve (Chapter 43) generally causes midfoot pain, but may present as medial ankle pain.

LATERAL ANKLE PAIN

Non-traumatic lateral ankle pain is generally associated with a biomechanical abnormality. The two most common diagnoses are peroneal tendinopathy and sinus tarsi syndrome. The causes of lateral ankle pain are listed in Table 42.3. The anatomy of the region is illustrated in Figure 42.8.

Examination

Examination is as for the patient with acute ankle injury (Chapter 41) with particular attention to testing resisted eversion of the peroneal tendons (Fig. 42.9a) and careful palpation for tenderness, swelling and crepitus (Fig. 42.9b).

Peroneal tendinopathy

The most common overuse injury causing lateral ankle pain is peroneus brevis tendinopathy. The peroneus longus and peroneus brevis tendons cross the ankle joint within a fibro-osseous tunnel, posterior to the lateral malleolus (Fig. 42.8a). The peroneus brevis tendon inserts into the tuberosity on the lateral aspect of the base of the fifth metatarsal. The peroneus longus tendon passes under the plantar surface of the foot to insert into the lateral side of the base of the first metatarsal and medial cuneiform. The peroneal tendons share a common tendon sheath proximal to the distal tip of the fibula, after which they have their own tendon sheaths. The peroneal muscles serve as ankle dorsiflexors in addition to being the primary evertors of the ankle.

Causes

Peroneal tendinopathy may occur either as a result of acute or recurrent ankle inversion injuries, or as an overuse injury. Soft footwear may predispose to the development of peroneal tendinopathy.

Common causes of an overuse injury to the peroneal tendons include:

- excessive eversion of the foot, such as occurs when running on a cambered surface⁷⁷
- excessive pronation of the foot
- secondary to tight ankle plantarflexors (most commonly soleus) resulting in excessive load on the lateral muscles
- excessive loading of the peroneals (e.g. dancing, basketball, volleyball).

An inflammatory arthropathy may promote peroneal tenosynovitis and subsequent peroneal tendinopathy. Peroneal tendinopathy may also be due to the excessive pulley action and abrupt change in direction of the peroneal tendons at the lateral malleolus. In some cases, peroneal tendinopathy is correlated with peroneal tendon instability in the retromalleolar groove of the lateral malleolus.

There are three main sites of peroneal tendinopathy:

1. posterior to the lateral malleolus (most common)
2. at the peroneal trochlea
3. at the plantar surface of the cuboid.

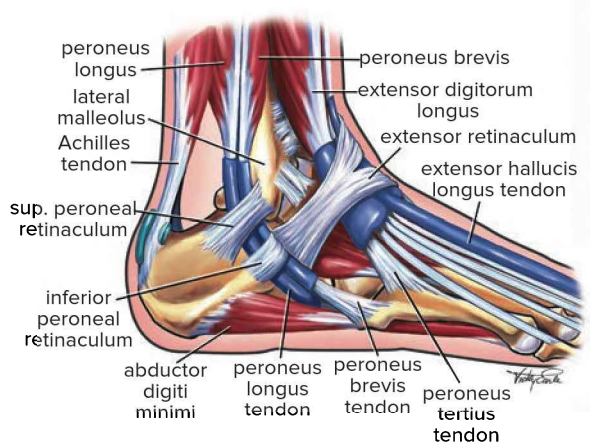
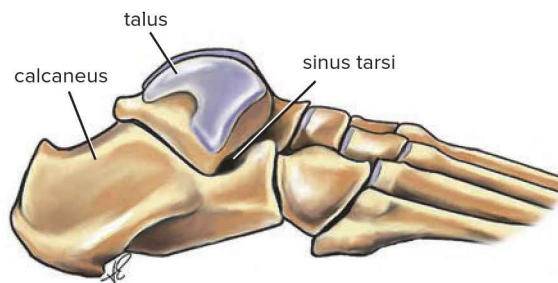
Clinical features

The athlete commonly presents with:

- lateral ankle or heel pain which is aggravated by activity and relieved by rest
- local tenderness of the peroneal tendons, sometimes associated with swelling and crepitus (a true paratenonitis)

Table 42.3 Causes of lateral ankle pain

Common	Less common	Not to be missed
Peroneal tendinopathy Sinus tarsi syndrome	Impingement syndrome • Anterolateral • Posterior Recurrent subluxation/dislocation of peroneal tendons Stress fracture of the talus Referred pain • Lumbar spine • Peroneal nerve • Superior tibiofibular joint	Stress fracture of the distal fibula Cuboid syndrome Complex regional pain syndrome type 1 (following knee or ankle trauma)

**(a)****(b)****Figure 42.8** Lateral aspect of the ankle (a) Anatomy of the lateral ankle (b) Sinus tarsi**(a)****(b)****Figure 42.9** Examination of the patient with lateral ankle pain (a) Resisted movement—eversion (peroneal muscles) (b) Palpation—the peroneal tendons are palpated for tenderness and crepitus

- painful passive inversion and resisted eversion, although in some cases eccentric contraction may be required to reproduce the pain
- calf muscle tightness
- excessive subtalar pronation or stiffness of the subtalar or midtarsal joints that is demonstrated on biomechanical examination.

Investigations

MRI is the recommended investigation and shows characteristic features of tendinopathy—increased signal and tendon thickening (Fig. 42.10)—and may show longitudinal tears.⁷⁸ If MRI is unavailable, ultrasound provides similarly good results. If an underlying inflammatory arthropathy is suspected, blood tests should be obtained to assess for rheumatological and inflammatory markers.

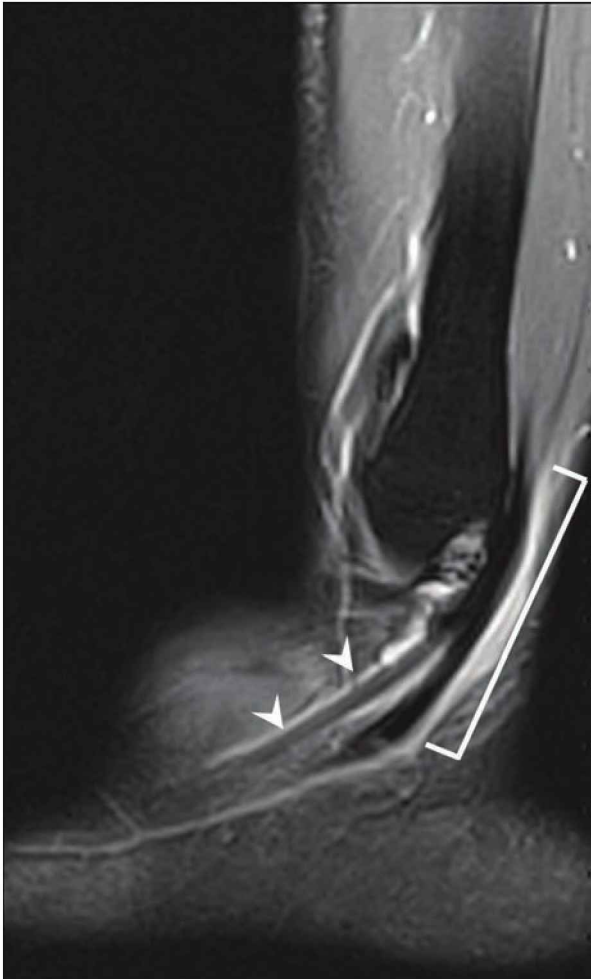


Figure 42.10 Peroneal tendinopathy. MRI showing oedema around the peroneus longus (bracket) and peroneus brevis (arrowheads)

Treatment

Treatment initially involves:

- rest from weight-bearing and loading activities; the use of a controlled ankle motion (CAM) boot may be required
- if a reactive tendinopathy is present, NSAIDs and regular icing
- soft tissue therapy to the peroneal muscles
- mobilisation of the ankle, subtalar and midtarsal joints
- footwear assessment: the use of lateral heel wedges or orthoses may also be required to correct biomechanical abnormalities
- strengthening exercises: should be introduced once the pain and inflammation has settled and should include resisted eversion (e.g. rubber tubing, rotagym), especially in plantarflexion as this position maximally engages the peroneal muscles.

PRP injections may be considered as an adjunct to a comprehensive rehabilitation program; however, there is currently little supporting evidence. In severe cases, or with failure of conservative management, surgery may be required, which may involve a synovectomy, tendon debridement, or repair of the tendon in case of longitudinal rupture.⁷⁹ If peroneal tendinopathy is associated with tendon instability, ligament reconstruction should address the instability of the tendon at the same time as the tendon is repaired.

Sinus tarsi syndrome

The sinus tarsi (Fig. 42.8b) is a small osseous canal running from an opening anterior and inferior to the lateral malleolus in a posteromedial direction to a point posterior to the medial malleolus. The interosseous ligament occupies the sinus tarsi and divides it into an anterior portion, which is part of the talocalcaneonavicular joint, and a posterior part, which represents the subtalar joint. It is lined by a synovial membrane and in addition to the ligament it contains small blood vessels, fat and connective tissue.

The sinus tarsi contains abundant synovial tissue that is prone to synovitis and inflammation when injured. An influx of inflammatory cells may result in the development of a low-grade and long-standing inflammatory synovitis. Although injury to the sinus tarsi may result from chronic overuse secondary to poor biomechanics (especially excessive pronation), approximately 70% of all patients with sinus tarsi syndrome have had a single or repeated inversion injury to the ankle. It may also occur after repeated forced eversion to the ankle, such as high jump take off. Other causes of sinus tarsi syndrome may include chronic inflammation in conditions such as gout, inflammatory arthropathies and osteoarthritis.

Clinical features

The symptoms of sinus tarsi syndrome include:

- pain which may be poorly localised and vague but is most often centred just anterior to the lateral malleolus
- pain that is often more severe in the morning and may diminish with exercise
- pain that may be exacerbated by running on a curve in the direction of the affected ankle; the patient may also complain of ankle and foot stiffness, a feeling of instability of the hindfoot and occasionally of weakness
- difficulty—often marked—walking on uneven ground
- full range of pain-free ankle movement on examination but the subtalar joint may be stiff
- pain on forced passive eversion of the subtalar joint; forced passive inversion may also be painful due to damage to the subtalar ligaments
- tenderness of the lateral aspect of the ankle at the opening of the sinus tarsi and occasionally also over the anterior talofibular ligament; there may be minor localised swelling.

Investigations

MRI may show an increased signal and fluid in the sinus tarsi, but is not often helpful. The most appropriate diagnostic test is injection of 1 mL of a short-acting local anaesthetic agent (e.g. 1% lignocaine [lidocaine]) into the sinus tarsi; ideally this should be done using fluoroscopy to ensure correct location (Fig. 42.11). In



Figure 42.11 Diagnostic local anaesthetic injection under fluoroscopic guidance. The needle is introduced into the lateral opening of the sinus tarsi with the foot in passive inversion. The needle should be directed medially and slightly posteriorly. Corticosteroid can be added if treatment required



Figure 42.12 Mobilisation of the subtalar joint is performed by medial-to-lateral transverse glide of the calcaneus on the talus

sinus tarsi syndrome, this injection will relieve pain so that functional tests, such as hopping on the affected leg, can be performed comfortably (for diagnosis). An ankle radiograph may be performed to exclude degenerative changes of the subtalar joint.

Treatment

Surgery is rarely indicated. Non-surgical management includes:

- relative rest
- short-term use of NSAIDs and ice
- mobilisation of the subtalar joint; this is essential (Fig. 42.12)
- direct infiltration of the sinus tarsi with corticosteroid (Fig. 42.11)
- correction of footwear and biomechanics
- proprioception and strength training.

Anterolateral impingement

Repeated minor ankle sprains or a major sprain involving the anterolateral aspect of the ankle may cause anterolateral impingement. An inversion sprain to the anterior talofibular ligament may promote synovial thickening. Usually this is subsequently resorbed, but sometimes this is incomplete and the residual tissue becomes hyalinised and moulded by pressure from the articular surfaces of the lateral aspect of the talus and lateral malleolus, where it may be trapped during ankle movements. A meniscoid lesion thus develops in the anterolateral gutter. Another postulated cause of anterolateral ankle impingement is chondromalacia of the lateral wall of the talus with an associated synovial reaction.

Clinical features

The classic presentation is pain at the anterior aspect of the lateral malleolus and an intermittent catching sensation in the ankle in an athlete with a previous history of ankle inversion injuries.

Examination may reveal tenderness in the region of the anteroinferior border of the fibula and anterolateral surface of the talus. End-range dorsiflexion, such as in a lunge position, is often reduced and painful. Proprioception may be poor.⁸⁰

Investigations

Clinical assessment is more reliable than MRI to diagnose this lesion.⁸¹ An arthroscopic examination confirms the diagnosis.

Treatment

In milder cases corticosteroid injection into the anterolateral gutter combined with physiotherapy and correction of biomechanics may be helpful. In more severe cases arthroscopic removal of the fibrotic, meniscoid lesion is required. Results after arthroscopic removal of the impinged tissue are encouraging and soccer players often return to sport after a short period (3–4 weeks) of rehabilitation. Surprisingly for a condition that limits dorsiflexion, it takes ballet dancers about 3 months to regain full pointe position (plantarflexion) after this operation.

Posterior impingement syndrome

Posterior impingement syndrome sometimes presents as lateral ankle pain, but more commonly as pain in the posterior ankle (Chapter 40).

Stress fracture of the talus

Stress fractures of the posterolateral aspect of the talus have been described in track and field athletes, triathletes, Australian rules footballers,⁸² elite female gymnasts⁸³ and a soccer player.⁸⁴

Causes

These stress fractures may develop secondary to excessive subtalar pronation and plantarflexion, resulting in impingement of the lateral process of the calcaneus on the posterolateral corner of the talus. In pole vaulters, this injury is usually acute and is attributed to 'planting' the pole too late.

Clinical features

The main symptom is that of lateral ankle pain of gradual onset, made worse by running and weight-bearing. Clinical



Figure 42.13 Stress fracture of the talus (arrowheads) demonstrated by coronal CT scan

examination reveals marked tenderness and occasionally swelling in the region of the sinus tarsi or posterior aspect of the ankle.

Investigations

Typical CT scan appearance is shown in Figure 42.13. MRI will also reveal the fracture, with the STIR sequence being most helpful (see Chapter 15).

Treatment

Treatment requires 6–8 weeks non weight-bearing immobilisation and then a supervised graduated rehabilitation. In elite athletes, when a rapid recovery is required, or in the case of failure of non-surgical management, surgical removal of the lateral process has been performed.

PRACTICE PEARL

As talar stress fractures are frequently associated with excessive pronation, biomechanical correction with orthoses is required before activity is resumed.

Referred pain

A variation of the slump test (Chapter 15) with the ankle in plantarflexion and inversion can be performed to detect increased mechanosensitivity in the peroneal nerve. If the test is positive, this position can be used as a glide or stretch in addition to soft tissue therapy to possible areas of restriction (e.g. around the head of the fibula).

ANTERIOR ANKLE PAIN

Pain over the anterior aspect of the ankle joint without a history of acute injury is usually due to either tibialis anterior tendinopathy or anterior impingement of the ankle. The surface anatomy of the anterior ankle is shown in Figure 42.14.

Anterior impingement of the ankle

Anterior impingement of the ankle joint (anterior tibiotalar impingement) is a condition in which additional soft or bony tissue is trapped between the tibia and talus during dorsiflexion. This may be the cause of chronic ankle pain or may result in pain and disability persisting after an ankle sprain. Although this syndrome has been called 'footballer's ankle', it is also seen commonly in basketballers and ballet dancers.

Anterior impingement occurs secondary to the development of exostoses (bone spurs) on the anterior rim of the tibia and on the upper anterior surface of the neck of talus (Fig. 42.15). The exostoses were initially described in ballet dancers and thought to be secondary



Figure 42.14 Surface anatomy of the anterior ankle showing tendons



Figure 42.15 X-ray showing bony exostosis on the anterior talus

to a traction injury of the joint capsule of the ankle that occurs whenever the foot was repeatedly forced into extreme plantarflexion. Subsequently, the development of the exostoses has been attributed to direct osseous impingement during extremes of dorsiflexion, as occurs with kicking in football and the *plié* (lunge) in ballet. As these exostoses become larger, they impinge on overlying soft tissue and cause pain.

Ligamentous injury and instability following ankle inversion injuries may also result in anterior ankle impingement; it has been shown that the distal fascicle of the anterior inferior tibiofibular ligament may impinge on the anterolateral aspect of the talus.⁸⁵

Clinical features

The patient complains of:

- anterior ankle pain, which initially starts as a vague discomfort
- pain ultimately becomes sharper and more localised to the anterior aspect of the ankle and foot, especially on dorsiflexion of the foot
- pain that is worse with activity, particularly with running, descending *plié* (lunge) in classical ballet, kicking in football or other activities involving dorsiflexion.

As the impingement develops, the patient complains of ankle stiffness, a limitation of *plié*, and in athletes a loss of take-off speed.

Examination reveals tenderness along the anterior margin of the talocrural joint. If the exostoses are



(a)



(b)

Figure 42.16 The anterior impingement test (a) The patient lunges forwards maximally and, if this reproduces the pain, the test is positive and suggests the diagnosis of anterior impingement (b) The same position is used to take a lateral X-ray. A positive test reveals bone-on-bone impingement (arrowed) when the patient adopts the lunge position that reproduces pain

large, they may occasionally be palpable. Dorsiflexion of the ankle is restricted and painful. The anterior impingement test (Fig. 42.16a), where the patient lunges forwards maximally with the heel remaining on the floor, reproduces the pain.

Investigations

Lateral X-rays in flexion and extension show both exostoses and abnormal tibiotalar contact. Ideally performed weight-bearing in the lunge position, bone-on-bone impingement confirms the diagnosis (Fig. 42.16b).

Treatment

In milder cases, non-surgical treatment consists of a heel lift, rest, modification of activities to limit dorsiflexion, NSAIDs and manual therapy, including accessory anteroposterior glides of the talocrural joint at the end of range of dorsiflexion. Taping or orthoses may help control the pain if they restrict ankle dorsiflexion or improve joint instability, which has been shown to contribute to the development of anterior impingement.⁸⁶

More prominent exostoses may require arthroscopic removal. The clinical results after arthroscopic removal are encouraging, and the majority of patients become pain-free with increased range of ankle motion.⁸⁷⁻⁸⁹

Tibialis anterior tendinopathy

The tibialis anterior tendon is the primary dorsiflexor of the foot, in addition to adducting and supinating (inverting) the foot. It passes medially over the anterior ankle joint and runs to insert into the medial and plantar aspects of the medial cuneiform bone and the adjacent base of the first metatarsal.

Causes

Tendinopathy of the tibialis anterior may result from overuse of the ankle dorsiflexors secondary to restriction in joint range of motion, as may occur with a stiff ankle. It may also be due to downhill running, playing racquet sports involving constant change of direction, or with excessive tightness of strapping or shoelaces over the tibialis anterior tendon.

Clinical features

The main symptoms are pain, swelling and stiffness in the anterior ankle. These symptoms are aggravated by activity, especially running, and walking up hills or stairs. On examination, there is localised tenderness, swelling and occasionally crepitus along the tibialis anterior tendon, especially over the anterior joint line. There is pain on resisted dorsiflexion and eccentric inversion. Long-standing and non-treated tendinopathy may eventually lead to partial or even total rupture of the tendon (under the extensor retinaculum).

Investigations

Ultrasound or MRI may be used to confirm the diagnosis and assess the severity of the tenosynovitis/tendinopathy.

Treatment

Treatment involves:

- rest from weight-bearing and loading activities
- the use of a CAM boot, if required

- if reactive tendinopathy present, NSAIDs and regular icing in the short term
- soft tissue therapy focusing on the tibialis anterior muscle in conjunction with ankle and subtalar joint mobilisation
- reducing running stride length; taking shorter steps decreases the functional length of the tibialis anterior and subsequently reduces the pull of the muscle on the tibia
- correction of biomechanics with the use of footwear and orthoses, if required

- eccentric strengthening and proprioception, required once the pain has settled.

In case of partial or total rupture, surgical reconstruction may be required.

Anterior inferior tibiofibular ligament (AITFL) injury

This injury is discussed in Chapter 41 because it results from an acute ankle injury (and often a fracture). If missed, it will present as persistent pain and loss of function after an ankle sprain.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e